Relation Algebra

§8. / is all of

(8a)

x (R/S) y⇔∀p. y S p→x R p

x (S\R) y⇔∀p. p S x→p R y

/compares images: S(y)⊆R(x). \ compares preimages: S°(x)⊆R°(y).

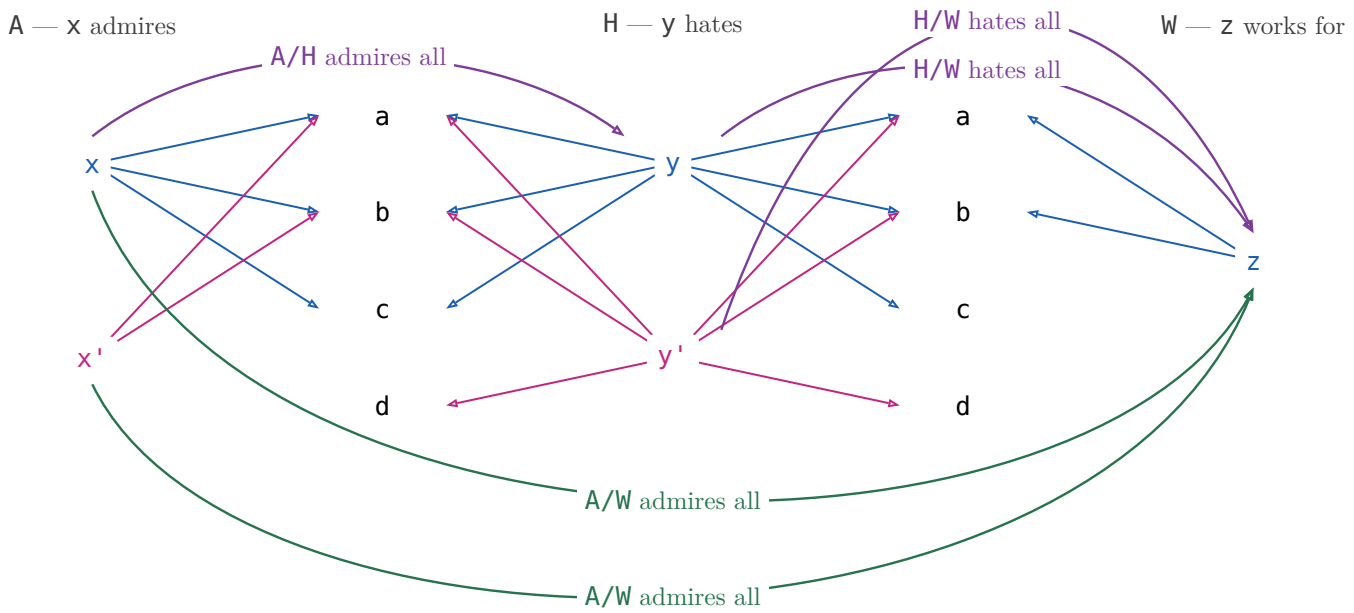
example: A admires,H hates,W works for

x (A/H) y — x admires everyone y hates.

x (H\A) y — everyone who hates x admires y.

§8.1. (R/S)(S/W)⊑R/W

(8.1a)



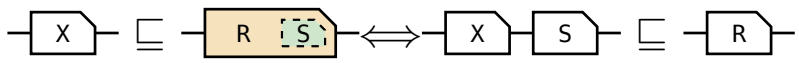
(8.1b)

x (A/H) y — x admires everyone y hates
y (H/W) z — y hates everyone z works for
x (A/W) z — x admires everyone z works for

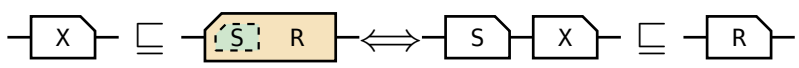
(A/H)(H/W) is a path: x → y → z, and that is all of it. A/W also holds of x', who admires everyone z works for — but x' does not admire everyone anybody hates, so nothing composes to it. The missing path is exactly the strictness of (R/S)(S/W)⊑R/W.

(8.1c)

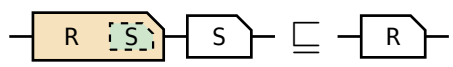
1 **X⊑R/S⇔XS⊑R**
X is any x-to-y pairing; one that only pairs x with a y such that x admires everyone y hates lies inside R/S, and R/S is the largest such.



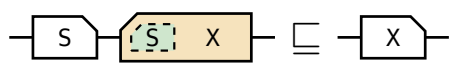
2 **X⊑S\R⇔SX⊑R**
The mirror — divide on the left when x comes first.



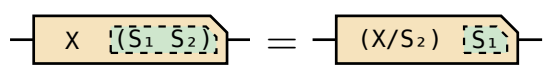
3 **(R/S)S⊑R**
There is a y such that x admires everyone y hates, and p is one of the people y hates — then x admires p too. Strict at S=∅: R/S is everyone, (R/S)S=∅.



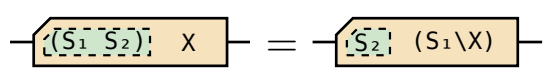
4 **S (S\R)⊑R**
The mirror.



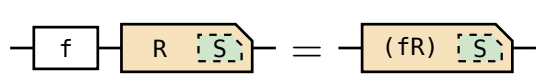
5 **associate: R/(S₁S₂)=(R/S₂)/S₁**
A friend's enemy is two hops: divide by the far end first.



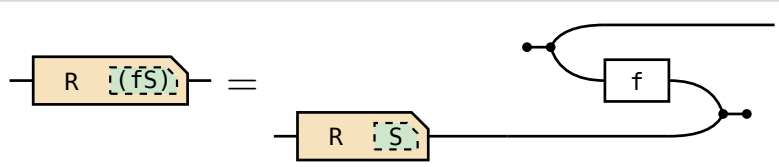
6 **(S₁S₂)\R=S₂\(S₁\R)**
The mirror.



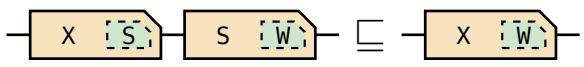
7 **maps: f (R/S)=(fR)/S**
Rename x before or after dividing — the licence to write fR/S.



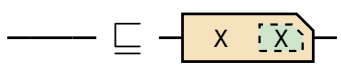
8 **R/(fS)=(R/S)f°**
Rename y: a map leaves a denominator as f° outside the box.



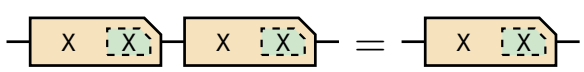
9 **(R/S)(S/W)⊑R/W**
Someone who admires all of a hate-set that already covers everyone z works for admires those people too.



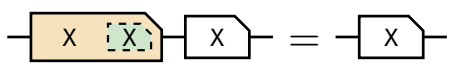
10 **1⊑R/R**
R/R runs admirer to admirer: each admires everyone they admire. Strict: two people who each admire only a and b admire each other's idols too, and still stay two people.



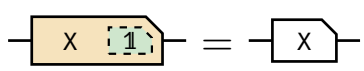
11 **(R/R)(R/R)=R/R**
R/R is the preorder **admires at least as much as**, and a preorder is idempotent. Freyd writes ⊑; with 1⊑R/R above it is an equality.



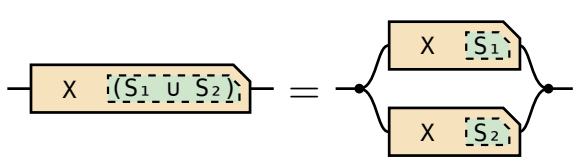
12 **(R/R)R=R**
Reaching p through someone whose idols x fully admires is reaching p directly, since x admires their own idols.



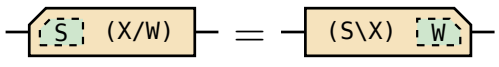
13 **R/1=R**
Dividing by 1: p's set is just {p}, so admiring all of it is admiring p.



14 **R/(S₁ ∪ S₂)=R/S₁∩R/S₂**
Admiring a combined hate-set is admiring each set in full.



15 **S\(R/W)=(S\R)/W**
Which is why S\R/W needs no bracket.



Fifteen laws, fifteen pictures, and not one shows a generator: ∩, ∪, ∘ and composition are what the Frobenius generators build, and / is none of those — it is posited, with nothing to unfold.